10-2. IPW and the Doubly Robust Estimator

A consolidated, mathematical lecture companion

English notes based on the lecture slides *IPW and Doubly Robust Estimator* (Prof. Sanghack Lee, GSDS SNU; slide credit Fan Li, Duke)

The central question is how to estimate $\mathbb{E}[Y(x)]$ and the ATE from observational data once unconfoundedness and positivity are granted, and how to protect the estimate against misspecification of the models we must fit.

Recall the key assumptions of the PO framework:

(1) unconfoundedness $\{Y(1), Y(0)\} \perp\!\!\!\perp X \mid Z$;

(2) positivity $0 < P(X = 1 \mid Z) < 1$;

(3) hence identification by adjustment, $\mathbb{E}[Y(x)] = \mathbb{E}_Z[\mathbb{E}[Y \mid X = x, Z]]$. These notes prove the two propensity-score properties, derive IPW and its normalized (Hájek) form, discuss trimming and balancing weights, and verify double robustness in both scenarios. Physical PDF page numbers are listed in Section 4.

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1 The propensity score

Purpose. Reduce the covariate vector to a scalar that suffices for confounding control.

Definition 1.1 (Propensity score; Rosenbaum and Rubin 1983)**.** $e(Z) = P(X = 1 \mid Z) = \mathbb{E}[X \mid Z]$, where $Z = (Z_1, \ldots, Z_p)$ collects the p pre-treatment covariates.

Proposition 1.2 (Property 1: balancing)**.** $X \perp\!\!\!\perp Z \mid e(Z)$. *More generally a* balancing score *is any $b(Z)$ with $X \perp\!\!\!\perp Z \mid b(Z)$; $e(Z)$ is one (Z itself trivially is another), and Rosenbaum and Rubin (1983) show that $e(Z)$ is the* coarsest *balancing score.*

Proof. It suffices to show $P(X = 1 \mid Z, e(Z)) = P(X = 1 \mid e(Z))$. The left side is $P(X = 1 \mid Z) = e(Z)$ since $e(Z)$ is a function of Z. The right side is $\mathbb{E}[\mathbb{E}[X \mid Z] \mid e(Z)] = \mathbb{E}[e(Z) \mid e(Z)] = e(Z)$. □

Proposition 1.3 (Property 2: unconfoundedness given the PS)**.** *If* $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid Z_i$, *then* $\{Y_i(1), Y_i(0)\} \perp\!\!\!\perp X_i \mid e(Z_i)$.

Proof. $P(X = 1 \mid Y(1), Y(0), e(Z)) = \mathbb{E}\big[P(X = 1 \mid Y(1), Y(0), Z) \mid Y(1), Y(0), e(Z)\big] = \mathbb{E}\big[e(Z) \mid Y(1), Y(0), e(Z)\big] = e(Z)$, which does not depend on the potential outcomes. □

Given covariates that ensure unconfoundedness, adjusting for differences in the propensity score removes all biases associated with differences in the covariates; $e(Z)$ is a summary score of the observed covariates.

2 Inverse probability weighting

Purpose. Estimate potential-outcome means by reweighting observed outcomes.

2.1 The IPW identity

$$\text{ATE} = \mathbb{E}\left[\frac{XY}{e(Z)} - \frac{(1-X)Y}{1-e(Z)}\right].$$

Using consistency ($XY = XY(1)$) and unconfoundedness ($X \perp\!\!\!\perp Y(1) \mid Z$),

$$\mathbb{E}\left[\frac{XY}{e(Z)}\right] = \mathbb{E}\left[\frac{\mathbb{E}[XY(1) \mid Z]}{e(Z)}\right] \overset{\text{uncf.}}{=} \mathbb{E}\left[\frac{\mathbb{E}[X \mid Z]\,\mathbb{E}[Y(1) \mid Z]}{e(Z)}\right] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid Z]\big] = \mathbb{E}[Y(1)],$$

and symmetrically for the control term. This is the Horvitz–Thompson (HT) estimator of survey sampling, also called inverse probability of treatment weighting (IPTW).

2.2 Estimators

With an estimated PS $\hat{e}$, define the weights $\hat{w}_1(Z_i) = 1/\hat{e}(Z_i)$ for $X_i = 1$ and $\hat{w}_0(Z_i) = 1/(1 - \hat{e}(Z_i))$ for $X_i = 0$. The difference of weighted means

$$\hat{\tau}_{\text{ipw},1} = \frac{1}{N}\left(\sum_{i=1}^{N} \frac{Y_i X_i}{\hat{e}(Z_i)} - \sum_{i=1}^{N} \frac{Y_i(1 - X_i)}{1 - \hat{e}(Z_i)}\right) = \frac{1}{N}\sum_{i=1}^{N} \{Y_i X_i \hat{w}_1(Z_i) - Y_i(1 - X_i)\hat{w}_0(Z_i)\}$$

is unbiased when $\hat{e} = e$.

2.3 Normalizing the weights

Since $\mathbb{E}\big[\mathbb{E}[X \mid Z]/e(Z)\big] = 1$, the weights average to one in expectation, but $\frac{1}{N}\sum_i X_i/\hat{e}(Z_i)$ need not equal one in a sample. Good practice for any weighting method is to normalize so that the weights within each group sum to one: divide $\hat{w}_i$ by $\sum_{j:X_j=x} \hat{w}_j$. This gives the Hájek estimator ("normalized" or "self-normalized" weights):

$$\hat{\tau}_{\text{ipw},2} = \frac{\sum_i Y_i X_i \hat{w}_1(Z_i)}{\sum_i X_i \hat{w}_1(Z_i)} - \frac{\sum_i Y_i (1-X_i)\hat{w}_0(Z_i)}{\sum_i (1-X_i)\hat{w}_0(Z_i)}.$$

It typically reduces variance, $V(\hat{\tau}_{\text{ipw},2}) \leq V(\hat{\tau}_{\text{ipw},1})$, and gives more stable estimates (Hirano, Imbens and Ridder 2003).

2.4 The weighted population

IPW creates a weighted (target) population that the sample represents. In it, covariate distributions are balanced between groups:

$$\mathbb{E}\left[\frac{X}{e(Z)}Z\right] = \mathbb{E}\left[\frac{1-X}{1-e(Z)}Z\right] \ (= \mathbb{E}[Z]),$$

since $\mathbb{E}[XZ/e(Z)] = \mathbb{E}[\mathbb{E}[X \mid Z]Z/e(Z)] = \mathbb{E}[Z]$. Intuition: the less likely a subject was to be treated, the larger the population it should represent.

2.5 Pros and cons

Advantages: simple, with a theoretical foundation (Horvitz–Thompson); model-free for the outcome; explicit target population (ATE, ATT, ...) via the choice of weights; extends to complex settings (time-varying treatments, marginal structural models).
Disadvantages: more sensitive to misspecification of the PS; propensity scores near 0 or 1 yield extreme weights, handled by many ad hoc decisions.

2.6 Trimming and truncation

- **Symmetric trimming** (Crump et al. 2009): exclude units whose estimated PS lies outside $[\alpha, 1-\alpha]$; rule of thumb $\alpha = 0.1$.
- **Asymmetric trimming** (Stürmer et al. 2010): exclude units outside the common PS range of treated and controls; then further exclude units whose PS is below the q quantile of the treated, and units whose PS is above the $(1-q)$ quantile of the controls.
- **Truncation:** set estimated PS below α to α and above $1-\alpha$ to $1-\alpha$.

Picture: the PS histograms of untreated (left-heavy) and treated (right-heavy) units have non-overlapping tails; range restriction removes the lowest percentiles of the treated and the highest percentiles of the untreated.

2.7 PS weighting in randomized trials

Although developed for observational studies, PS weighting can be used for covariate adjustment in RCTs: chance imbalances are common, and adjustment improves precision. The procedure is the same as in observational studies (with an estimated, rather than the known, PS).

2.8 Weighting versus balancing

IPW balances covariates *in expectation* but not in finite samples. A recent line of work finds weights w_i directly so that weighted covariate means are (approximately) balanced, bypassing PS estimation:

$$\frac{1}{N_1}\sum_{i=1}^{N} w_i X_i Z_i \approx \frac{1}{N_0}\sum_{i=1}^{N} w_i (1-X_i) Z_i,$$

typically minimizing a measure of weight dispersion (e.g. entropy) subject to balance constraints, by numerical optimization. Examples: entropy balancing (Hainmueller 2011), covariate balancing propensity score (Imai and Ratkovic 2014), stabilized balancing weights (Zubizarreta 2015). These balance only the functions (moments) of Z the analyst specifies. If the true outcome model is $Y \sim Z_1 + Z_2 + Z_1 Z_2$ and we balance $Z_1, Z_2, Z_1^2, Z_2^2, \ldots$ but not $Z_1 Z_2$, the interaction may be severely imbalanced and the result biased.

3 Doubly robust estimation

Purpose. Combine outcome regression and IPW so that consistency requires only one of the two models to be correct.

3.1 Terminology

- **Correctly specified model:** the true function belongs to the assumed class (e.g. $\mathbb{E}[Y \mid X, Z] = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ$ fitted with a linear model with interaction).
- **Misspecified model:** it does not (e.g. the truth is non-linear in Z but a linear model is fitted). The fit converges to the best approximation within the class, which is generally biased.
- **Consistent estimator:** $\hat{\theta}_N \xrightarrow{p} \theta$ as $N \to \infty$. Correct specification gives consistency (under regularity conditions); misspecification generally does not.

3.2 Two building blocks

Since $\tau = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)]$ is symmetric under $X \leftrightarrow 1-X$, $e \leftrightarrow 1-e$, focus on $\mu_1 = \mathbb{E}[Y(1)]$. Let $m_x(Z) = \mathbb{E}[Y(x) \mid Z]$ (written $\mu_x(z)$ in Lecture 10-1).

- **Outcome regression:** $\hat{\mu}_{1,\text{adj}} = N^{-1}\sum_i \hat{m}_1(Z_i)$; consistent if m_1 is correctly specified.
- **IPW:** $\hat{\mu}_{1,\text{ipw}} = N^{-1}\sum_i Y_i X_i / \hat{e}(Z_i)$; consistent if the PS model (e.g. logistic) is correct, but can be inefficient.

Can the virtues be combined? Yes:

$$\hat{\mu}_{1,\text{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right].$$

3.3 Verification with the true nuisance functions

$$\mathbb{E}\left[\frac{XY}{e(Z)} - \frac{X-e(Z)}{e(Z)}m_1(Z)\right] = \mathbb{E}\left[\frac{XY - Xm_1(Z) + e(Z)m_1(Z)}{e(Z)}\right] = \mathbb{E}\left[m_1(Z) + \frac{X}{e(Z)}\{Y - m_1(Z)\}\right] = \mu_1,$$

since by iterated expectations and $\mathbb{E}[X \mid Z] = e(Z)$ (as in the IPW derivation) $\mathbb{E}[X\{Y(1) - m_1(Z)\}/e(Z)] = \mathbb{E}[\mathbb{E}[Y(1) - m_1(Z) \mid Z]] = 0$.

3.4 Two equivalent forms

$$\hat{\mu}_{1,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right] \quad \text{(IPW augmented by OR)}$$
$$= \frac{1}{N}\sum_{i=1}^{N}\left[\hat{m}_1(Z_i) + \frac{X_i\{Y_i - \hat{m}_1(Z_i)\}}{\hat{e}(Z_i)}\right] \quad \text{(OR augmented by IPW residual correction).}$$

The forms are algebraically identical but suggest different statistical interpretations.

3.5 What does it estimate?

With arbitrary limits $\hat{e}$, $\hat{m}_1$, using $XY = XY(1)$:

$$\mathbb{E}\left[\frac{XY}{\hat{e}(Z)} - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right] = \mathbb{E}\left[Y(1) + \frac{X - \hat{e}(Z)}{\hat{e}(Z)}Y(1) - \frac{X - \hat{e}(Z)}{\hat{e}(Z)}\hat{m}_1(Z)\right]$$
$$= \mathbb{E}[Y(1)] + \underbrace{\mathbb{E}\left[\frac{X - \hat{e}(Z)}{\hat{e}(Z)}\{Y(1) - \hat{m}_1(Z)\}\right]}_{R}.$$

$\hat{\mu}_{1,\mathrm{dr}}$ estimates $\mathbb{E}[Y(1)]$ iff $R = 0$: a regression estimator augmented by weighted residuals.

Scenario 1: PS wrong, OR right ($\hat{m}_1 = m_1$). Conditioning on Z and using unconfoundedness,

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right], \qquad \mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z] = m_1(Z) - \hat{m}_1(Z) = 0.$$

Scenario 2: PS right, OR wrong ($\hat{e} = e$).

$$R = \mathbb{E}\left[\frac{e(Z) - \hat{e}(Z)}{\hat{e}(Z)}\,\mathbb{E}[Y(1) - \hat{m}_1(Z) \mid Z]\right] = 0 \quad \text{since } e(Z) - \hat{e}(Z) = 0.$$

In both cases $R = 0$ and $\hat{\mu}_{1,\mathrm{dr}}$ is consistent. By the same argument with $X \to 1 - X$, $e \to 1 - e$, $m_1 \to m_0$,

$$\hat{\mu}_{0,\mathrm{dr}} = \frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]$$

is consistent for $\mathbb{E}[Y(0)]$, and if both models are correct it is of course consistent too.

3.6 The doubly robust ATE estimator

$$\hat{\tau}_{\mathrm{dr}} = \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{X_i Y_i}{\hat{e}(Z_i)} - \frac{X_i - \hat{e}(Z_i)}{\hat{e}(Z_i)}\hat{m}_1(Z_i)\right]}_{\hat{\mu}_{1,\mathrm{dr}}}$$
$$- \underbrace{\frac{1}{N}\sum_{i=1}^{N}\left[\frac{(1 - X_i)Y_i}{1 - \hat{e}(Z_i)} + \frac{X_i - \hat{e}(Z_i)}{1 - \hat{e}(Z_i)}\hat{m}_0(Z_i)\right]}_{\hat{\mu}_{0,\mathrm{dr}}}.$$

3.7 Concluding remarks

- **Double robustness:** $\hat{\tau}_{\text{dr}}$ is consistent if either the PS model or the outcome model (not necessarily both) is correctly specified: two chances to get it right. DR is a large-sample property.
- If both e and m_x are correct, $\hat{\tau}_{\text{dr}}$ has smaller asymptotic variance than IPW. If the outcome model is correct, it has larger asymptotic variance than the direct regression estimator.
- Empirically the outcome model matters more: misspecifying $\hat{m}$ causes much larger bias than misspecifying $\hat{e}$, and even with a correct $\hat{e}$ a misspecified $\hat{m}$ usually gives notable bias; the extra protection from PS augmentation is somewhat limited.
- Kang and Schafer (2007, *Stat. Sci.*) give an example where moderate misspecification of both models (a likely scenario) leads to large bias and variance of DR.

3.8 Summary

1. The propensity score $e(Z) = P(X = 1 \mid Z)$ is a scalar summary of covariates that balances treatment groups.
2. IPW reweights by $1/e(Z)$ to remove confounding. Pros: simple, outcome-model-free. Cons: extreme weights.
3. Outcome regression models $\mathbb{E}[Y \mid X, Z]$ directly. Pros: efficient when correct. Cons: sensitive to misspecification.
4. Doubly robust estimation combines both and is consistent if either model is correct.

4 Source coverage

The source deck has 40 physical pages (31 numbered slides).

Topic	Physical pages
Credit, recall, propensity score and its properties	1–7
IPW identity, estimators, normalization, weighted population	8–16
Pros/cons, trimming/truncation, RCTs, balancing weights	17–24
Doubly robust estimation, scenarios, remarks, summary	25–40